Linear Programming Course Summary

Generated by Scribe.AI

December 4, 2024

1 Key Terms

1.1 linear programming fundamentals

- **Slack Variable**: *A variable added to a less-than-or-equal-to inequality constraint to transform it into an equality constraint. The slack variable represents the difference between the left-hand side and the right-hand side of the inequality.*

1.2 solution space analysis

- **Feasible Region**: *The set of all points that satisfy all the constraints of a linear programming problem.*
- **Vertex**: *A point where two or more constraint boundaries intersect in the feasible region of a linear programming problem. In the Simplex method, optimal solutions are found at vertices.*

1.3 simplex method algorithms

- **Basic Variable**: *In the Simplex method, a variable that is expressed in terms of non-basic variables in a dictionary. Basic variables are set to values that satisfy the constraints.*
- **Non-basic Variable**: *In the Simplex method, a variable that is set to zero in a given iteration of the algorithm. The values of the basic variables are determined by the values of the non-basic variables.*
- **Pivot Operation**: *The process of exchanging a basic variable with a non-basic variable in the Simplex method, leading to a new dictionary and a movement to a different vertex in the feasible region.*
- **Dictionary**: *A representation of a linear programming problem in a tabular format, where basic variables are expressed as functions of non-basic variables. It is used in the Simplex method to systematically move from one vertex to another.*
- **Auxiliary Problem**: *A modified linear programming problem introduced to find an initial feasible solution when the origin is not feasible in the original problem. It involves adding a new variable to shift the feasible region.*

2 Problem Types

2.1 linear programming fundamentals

- **Optimizing with Slack Variables**: *The Simplex method uses slack variables to convert inequality constraints into equalities, facilitating the iterative optimization process.*
- **Graphical Representation of Feasible Regions**: *Visualizing the feasible region (the set of points satisfying all constraints) helps understand the Simplex method's iterative movement between vertices.*
- **Solving Linear Programs with Three Variables**: *Extending the Simplex method to problems with more than two variables requires careful consideration of the constraints and iterative optimization.*

2.2 advanced linear programming techniques

- **Finding an Initial Feasible Solution**: *The Simplex method requires an initial feasible solution. If the origin is not feasible, an auxiliary problem is introduced to find a feasible solution to the original problem.*
- **Handling Infeasible Origins**: *The origin* $(0, 0)$ *may not satisfy all constraints in a linear programming problem. Techniques are needed to find a feasible starting point for the Simplex method.*
- **Pivot Operations in the Simplex Method**: *The Simplex method iteratively improves the solution by exchanging basic and non-basic variables (pivot operations) in the dictionary.*
- **Working with Dictionaries of Variables**: *The Simplex method uses a dictionary to represent the linear programming problem, where basic variables are expressed in terms of non-basic variables. Each dictionary corresponds to a vertex of the feasible region.*

3 Algorithm Solutions

3.1 linear optimization techniques

- **Simplex Method**: *Iteratively move from one vertex of the feasible region to another by setting non-basic variables to zero and choosing a new set of (non)basic variables to improve the objective function value. This is equivalent to moving from vertex to vertex in the feasible region.*
- **Auxiliary Problem**: *When the origin is not feasible, introduce a new variable* x_0 *and modify the constraints to create an auxiliary problem that maximizes* $-x_0$*. Solve this auxiliary problem to find a feasible solution for the original problem. If the optimal solution has* $x_0 = 0$*, then a feasible solution to the original problem is found.*